FunnelDoctor AI

Master Engineering Roadmap & Prioritized Backlog

DOCUMENT: FD-ROADMAP-V2
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CURRENT STATUS	COMPLETED GATES	TOTAL BACKLOG ITEMS	EST. COMPLETION COST
223 / 223	Gates 1 - 5	8 Tasks	$0.00
Tests Passing (100% Green)	Milestones 1 & 2 Complete	Prioritized P0 to P3	Uses Free Tiers & Open APIs

Executive Strategic Context: FunnelDoctor AI has completed all core scientific, cryptographic, and diagnostic foundations (Milestones 1 & 2). This roadmap establishes the chronological and prioritized backlog required to transition from local/in-memory proof to an automated, self-serve, multi-tenant enterprise CRO platform.

Priority 0 (P0) — Immediate Cloud Persistence & Field Verification

TIMELINE: DAYS 1 - 2

1.1 Implement PostgreSQL / Supabase Driver (PostgresDatabaseAdapter)

P0 BLOCKER | Est: 3 Hours | Cost: $0.00

Build concrete `PostgresDatabaseAdapter` implementing the universal `DatabaseAdapter` interface using the high-performance `postgres` (or `pg`) driver. Enables seamless automatic switching from local SQLite to cloud-persistent Supabase PostgreSQL in Vercel serverless environments via `DATABASE_URL`.

Files: `packages/core/src/database/postgres-adapter.ts`, `packages/core/src/database/index.ts`

Technical Specs: SSL mode required (`sslmode=require`), parameterized queries ($1, $2), atomic connection pooling.

Dependencies: `pnpm --filter @funneldoctor/core add postgres`

Deliverable: Cloud database retains merchant sessions, events, and audit logs permanently across serverless reboots.

1.2 Wire Managed Upstash Redis Cluster in Production

P0 CRITICAL | Est: 1 Hour | Cost: $0.00

Connect production `REDIS_URL` to `RedisStateStore`. Enforces multi-region distributed rate-limiting (120 req/min/IP), atomic webhook idempotency locks (`setNX`), and sliding-window Lua script evaluation to protect the ingestion API.

Files: `apps/web/.env.production`, `packages/core/src/state-store.ts`

Technical Specs: REST/HTTP fallback for Upstash Serverless Redis, TLS encryption in transit.

Dependencies: Free Upstash Redis instance (10,000 commands/day free).

Deliverable: Production Vercel environment guarded against DDoS, telemetry flooding, and duplicate webhooks.

1.3 Live Storefront Field Verification Protocol

P0 QUALITY | Est: 2 Hours | Cost: $0.00

Execute the formal 10-point field verification workbook on a live Shopify development store. Verify zero console errors, zero layout shift (CLS = 0.00), Safari ITP cookie independence, and fail-safe beacon transmission on checkout close.

Files: `docs/FIELD_VERIFICATION_CHECKLIST.md`

Devices Tested: iOS 17 Safari, Android 14 Chrome, Instagram In-App WebView.

Dependencies: Free Shopify Partner development store.

Deliverable: Signed verification log proving storefront safety before merchant onboarding.

Priority 1 (P1) — 1-Click Store Integrations & Zero-Code Ingestion

TIMELINE: WEEK 1

2.1 Shopify OAuth 2.0 Handshake Flow (/api/auth/shopify)

P1 EXPERIENCE | Est: 6 Hours | Cost: $0.00

Implement automated Shopify App Bridge OAuth exchange. Merchants enter their store handle (e.g. `store.myshopify.com`) and click "Connect". Shopify presents the permissions grant dialog (`read_orders`, `read_checkouts`, `read_products`), and FunnelDoctor securely exchanges the code for a permanent offline access token.

Files: `apps/web/src/app/api/auth/shopify/route.ts`, `apps/web/src/app/api/auth/shopify/callback/route.ts`

Security: Nonce verification against CSRF; AES-256-GCM envelope encryption with AAD tenant binding.

Dependencies: Free Shopify Partner App credentials (Client ID / Secret).

Deliverable: 1-Click store connection without merchants touching code or API keys.

2.2 Shopify GraphQL Historical Data Sync Engine

P1 INSTANT VALUE | Est: 5 Hours | Cost: $0.00

Immediately upon OAuth completion, a background sync worker queries Shopify GraphQL Admin API for the last 90 days of completed orders, revenue, AOV, and abandoned checkouts. Populates `aggregated_funnel_metrics` within 15 seconds so the dashboard displays 100% exact historical data.

Files: `packages/core/src/shopify-sync.ts`, `apps/web/src/app/api/sync/shopify/route.ts`

API Query: `orders(first: 250, query: "created_at:>90d")`, `abandonedCheckouts(first: 250)`

Rate Limits: Shopify GraphQL Cost calculation (leaky bucket algorithm, 50 points/sec max).

Deliverable: Merchants see 90 days of exact revenue leakage and historical drop-offs in 15 seconds.

2.3 Automated ScriptTag / Web Pixel API Auto-Injection

P1 ZERO-CODE | Est: 3 Hours | Cost: $0.00

Programmatically inject the 2.8KB FunnelDoctor tracker via Shopify's official `ScriptTag API` or `Web Pixels API`. Eliminates manual Liquid editing; the merchant is fully instrumented the moment they authorize the app.

Endpoint: `POST /admin/api/script_tags.json`

Configuration: `event: "onload"`, `src: "https://funneldoctorai.vercel.app/tracker.js"`

Safety: Asynchronous defer load, SRI hash integrity verification.

Deliverable: Zero-friction installation; merchant never opens theme code.

2.4 Outbound Slack & WhatsApp Anomaly Alert Dispatcher

P1 RETENTION | Est: 4 Hours | Cost: $0.00

Implement webhook alert engine that notifies merchants via Slack or WhatsApp Business API when checkout drop-off spikes abnormally (e.g. payment failure rate exceeds 20% within 1 hour due to UPI gateway outage).

Files: `apps/web/src/app/api/alerts/route.ts`, `packages/core/src/alert-dispatcher.ts`

Channels: Slack Incoming Webhooks, Resend Email API, Gupshup/Twilio WhatsApp REST API.

Deduplication: 4-hour alert cooling window to prevent notification spam.

Deliverable: Instant revenue protection; alerts merchants before customers file support tickets.

Priority 2 (P2) — Self-Serve Monetization & Billing

TIMELINE: WEEK 2 – 3

3.1 Automated Stripe & Razorpay Subscription Billing Portal

P2 REVENUE | Est: 8 Hours | Cost: Standard 2% Fee

Build self-serve billing portal with dual-currency support (USD on Stripe, INR on Razorpay Subscriptions). Features Growth Tier ($199/mo | ₹14,999/mo) and Scale Tier ($499/mo | ₹39,999/mo). Webhook listener handles automated tenant provisioning, grace periods, and cancellations.

Files: `apps/web/src/app/api/billing/webhook/route.ts`, `apps/web/src/app/pricing/page.tsx`

Capabilities: Automatic GST tax invoices, Razorpay Recurring UPI / Auto-debit (Mandate compliant).

Priority 3 (P3) — Ecosystem Distribution & Scaled Partnerships

TIMELINE: MONTH 2

4.1 Official Shopify App Store Submission & Listing Package

P3 DISTRIBUTION | Est: 12 Hours | Cost: $0.00

Package FunnelDoctor as an embedded Shopify App using Shopify App Bridge and Polaris design system. Implement mandatory GDPR/DPDP privacy webhooks (`customers/data_request`, `customers/redact`, `shop/redact`) and submit for official Shopify Partner review.

Review Requirements: App Bridge v3, OAuth session tokens, Shopify App Billing API integration.

Distribution Impact: Access to over 2,000,000 active Shopify merchants worldwide.

Strategic Positioning: FunnelDoctor vs. Industry Landscape

Platform	Core Job-to-be-Done	What It Does NOT Do	FunnelDoctor Advantage
Triple Whale ($129–$999/mo)	Ad attribution & blended ROAS tracking (Meta/Google ad spend cockpit).	Does NOT track checkout DOM friction, rage clicks, or write code fixes.	FunnelDoctor diagnoses *why* checkouts fail and writes the code to fix the leak.
Lifetimely ($49–$399/mo)	LTV calculation, cohort repurchase rates, and net profit accounting.	Does NOT diagnose storefront friction or optimize real-time conversion.	FunnelDoctor recovers lost first-purchase revenue before LTV even begins.
Elevar ($250–$1,200/mo)	Server-side data plumbing (Meta CAPI & GA4 server tags).	Zero diagnostic UI, zero revenue advice, zero code remediation.	FunnelDoctor provides actionable diagnostic intelligence with ready-to-deploy code.
Mixpanel / Amplitude (Usage-based)	General software product analytics for apps, SaaS, and games.	Zero out-of-the-box Shopify integration; requires custom engineers.	FunnelDoctor is purpose-built for e-commerce with built-in UPI, COD, and AOV math.

Master Priority Execution Schedule

Phase	Priority	Sprint Target	Key Deliverable	Resource Cost
Phase 1	P0	Days 1 – 2	Postgres (Supabase) Adapter + Redis + Live Dev Store Field Verification	$0.00 (Free Tiers)
Phase 2	P1	Week 1	1-Click Shopify OAuth + 15s Historical GraphQL Sync + Auto-Pixel Injection	$0.00 (Partner API)
Phase 3	P2	Week 2 – 3	Self-Serve Stripe/Razorpay Billing + Outbound Slack/WhatsApp Alerts	Standard 2% Fee
Phase 4	P3	Month 2	Official Shopify App Store Submission + WooCommerce Plugin Distribution	$0.00 (Partner Acct)